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Detection and classification of enhanced periapical lesion images with YOLO algorithms

Fatma Akalin^a and Tuğçenur Yıldız^b

^aDepartment of Information Systems Engineering, Faculty of Computer and Information Sciences, Sakarya University, Sakarya, Turkey; ^bDepartment of Endodontics, Faculty of Dentistry, Sakarya University, Sakarya, Turkey

ABSTRACT

In recent years, artificial intelligence has become a reliable technology in clinical decision support systems with the solutions it offers in the dental field. This success also indicates a significant potential in detecting five different types of lesions on the tooth root through radiographic images. Because the periapical X-ray-taking process may vary depending on the individual's physical, psychological, and mental conditions. Also, environmental parameters may negatively affect the image acquisition process. It is possible to tolerate these disadvantageous situations with artificial intelligence-based algorithms and image processing approaches. In this study, it was planned to in-depth analysis of the periapical lesion data types in the original dataset called Periapical X-rays provided from the Kaggle public database. For this, the original adaptive image processing approach developed by integrating the ABC optimisation algorithm was applied to the dataset for five different lesion types. Then, the enhanced images enriched with the data augmentation approach were trained with YOLOv7, YOLOv8, YOLOv9 and YOLOv10 algorithms. As a result of the training, the enhanced images compared to the original images reached 96% F Criterion thanks to the network architecture of YOLOv8 algorithm. This shows that the YOLOv8 network architecture in enhanced lesion images is more successful.

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1. Introduction

Teeth are structures designed to bite and break food into small pieces. It enables digestive enzymes to interact effectively with food grains (Johnson, 2011). It is important for the physiology and beauty of the face. It also ensures the correct pronunciation of words and the protection of supporting organs. In order for teeth to perform all these functions, it is critical that they have a good anatomical shape and good health (Chen & Liu, 2014).

Healthy teeth have two types of tissue: hard and soft. The hard tissue has a calcified structure. It refers to enamel, dentin, and cementum. The soft tissue is located in the centre of the tooth. It refers to the pulp rich in blood vessels and nerves (Chen & Liu, 2014). The pulp is connected to the surrounding tissues of the tooth (periodontal tissues) through a hole at

CONTACT Fatma Akalin ✉ fatmaakalin@sakarya.edu.tr Department of Information Systems Engineering, Faculty of Computer and Information Sciences, Sakarya University, Sakarya, Turkey

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the apex of the root, through dentin tubules or lateral canals (Torabinejad & Rotstein, 2014). Therefore, the pulp is anatomically very closely related to the periodontal tissues.

Deterioration of dental health can be caused by pulp and/or periodontal tissues. The development of pulp-derived disease usually begins with tooth decay. The process of tooth decay first occurs in the hard tissues of the teeth. This formation occurs by chemical dissolution as a result of the activity of microbial accumulation on the surface of the tooth enamel (Kidd & Fejerskov, 2016). If the dissolution is left untreated, hard tissues such as dentin and enamel will be damaged, as well as the inner part containing the pulp (Sajad et al., 2019). As a result of the anatomical neighbourhood, periodontal tissue and bone damage will be inevitable over time (Li et al., 2022). The development of a periodontal-related disease is a condition that begins in the surrounding tissues of the tooth and affects the dental pulp if left untreated. As a result of this condition, the inflammation of the surrounding tissues of the tooth is called periodontitis, and the bone resorption caused by inflammation is called lesion. These lesions are classified according to their location on the tooth (the apical of the root, lateral part of the root, part of the tooth close to the crown, etc.). Endo-perio lesions are primary endodontic lesion, primary periodontal lesion, primary endodontic lesion with secondary periodontal involvement, primary periodontal lesion with secondary endodontic involvement and true combined lesions (Simon et al., 2013).

The symptoms of endo-perio lesions are often similar due to the close anatomical and functional relationship between the pulp and periodontal tissues (Dakó et al., 2020; Evans, 2023). Periapical x-rays of these types of lesions, which complicate the diagnostic process for dentists (Al-Fouzan, 2014; Berman & Hargreaves, 2020; Dakó et al., 2020; Torabinejad & Rotstein, 2014), are shown in Figure 2. The types of endo-perio lesions are:

- 1) Primary endodontic lesions: Inflammatory alterations are often a part of dental pulp disease processes. The most frequent causes include dental caries, restorative procedures, and traumatic injuries. Endodontic lesions usually cause the attachment apparatus next to a nonvital tooth to be destroyed as well as resorb bone apically and laterally (Berman & Hargreaves, 2020).
- 2) Primary periodontal lesions: It is well known that periodontal disease progresses to the formation of bone defects. These defects often appear radiographically along the lateral surfaces of the roots and in the furcation areas (Berman & Hargreaves, 2020).
- 3) Primary endodontic lesions with secondary periodontal involvement: If an endodontic lesion is left untreated, the disease usually worsens, destroying the periapical alveolar bone and spreading into the interradicular space, rupturing the surrounding soft and hard structures (Berman & Hargreaves, 2020).
- 4) Primary periodontal lesions with secondary endodontic involvement: The temporal order of the disease processes is the only distinction between primary periodontal lesions with endodontic involvement and primary endodontic lesions with secondary periodontal involvement (Berman & Hargreaves, 2020). On radiography, the angular bone loss in this type of lesion extends from the cervical region to the apex (Torabinejad & Rotstein, 2014).
- 5) True combined lesions: Pulpal and periodontal diseases can develop in and around the same tooth separately or together (Berman & Hargreaves, 2020). When an infected periodontal pocket that is growing apically meets an endodontic illness

that is progressing coronally, it results in true combined endo-perio disorders. In this kind of injury, attachment loss is quite severe (Torabinejad & Rotstein, 2014).

The order of occurrence and differentiation of disease in pulp and periodontal tissue structures, which are the origin of the classification and are anatomically closely related, is quite difficult. In evaluating this complex situation, the patient's medical history, vitality of the tooth, presence of gingival pockets, mobility of the teeth and dental radiographs should be carefully examined (Berman & Hargreaves, 2020). In particular, dental periapical radiographs are one of the most commonly used tools for the diagnosis of diseases in endodontic practice (Gliga et al., 2023; Shafi et al., 2023b). Accessibility, ease of operation, low cost and limited radiation exposure are important reasons for choosing this method (Xambre et al., 2016). In addition to these advantages, there are also a number of limitations. First of all, the anatomy of the tooth and surrounding structures, which are three-dimensional, is compressed into a two-dimensional image (Gröndahl & Huumonen, 2004). In addition, environmental factors such as low image contrast, less brightness and the presence of noise, etc. reduce image quality (Rahmi-Fajrin et al., 2018). Due to all these parameters, it becomes difficult for professionals to separate the X-ray into different regions and get the necessary information with their naked eyes. Moreover, human examination leads to subjective judgments. These subjective judgments may be influenced by environmental disadvantages affecting the image, as well as by the physical, social and mental changes of the examining professional. These factors can lead professionals to misinterpret radiographs (Muresan et al., 2020). The manual process of radiograph evaluations, especially by inexperienced dentists, can be tedious and error-prone (Fatima et al., 2023; Shafi et al., 2023a). This situation may have a negative impact on the diagnostic period (Li et al., 2006). On the other hand, some early bone loss and changes are at a level that cannot be seen with human eyes (Li et al., 2006). As with many diseases, early diagnosis of bone loss is critical. Because if early detection is not provided, tooth loss may occur (Li et al., 2006).

Medical decision making is difficult. The situation becomes difficult especially at times when it is necessary to perform procedures on a large number of patients within a narrow time parameter. It would be beneficial to develop an computer-aided tool to support doctors in making accurate and rapid decisions (Shafi et al., 2021). Artificial intelligence-based decision support systems can be used to these all situations. The use of this technology is essential, especially for accurate and early diagnosis. It has the potential to strengthen the patient management of physicians (Abdalla-Aslan et al., 2020).

Artificial intelligence is a framework that enables human cognitive abilities to be realised by machines (Hashimoto et al., 2020). It allows manual processes to be managed faster and easier, regardless of environmental parameters (Akalin & Yumusak, 2022a). It is used in cases of being able to perform pattern completion, self-organising, working with incomplete information, having fault tolerance, showing graded distortion, solving nonlinear problems and producing answers to situations that do not have a mathematical model (Ümit Aksoylu, 2021). Artificial intelligence has been widely used in the field of dentistry in recent years. Fatima et al. (Fatima et al., 2022) have highlighted the current advances in integrating artificial intelligence, machine learning and deep learning in the subfields of dentistry, including periodontology, endodontics, orthodontics, restorative dentistry and oral pathology. According to this study, they concluded that artificial intelligence has the potential to improve dental care, disease diagnosis and prognosis, treatment planning, and

risk assessment. Muramatsu et al. (Muramatsu et al., 2021) have aimed to train an object detection network using the quadruple cross-validation method in their study in which they developed a computer-aided system to detect and classify teeth in panoramic radiographs. As a result, the tooth detection sensitivity was 96.4% with 0.5 false positives per case. The classification accuracy for tooth types and dental conditions was reported as 93.2% and 98.0%. They stated that by using a double input layer network, a 6-point increase in classification accuracy for tooth types was achieved. Lee et al. (Lee et al., 2021) have tested the reliability and validity of three different deep convolutional neural network (DCNN) architectures (VGGNet-19, GoogLeNet Inception-v3, and auto DCNN) for the detection and classification of fractured dental implants using panoramic and periapical radiographic images. As a result, they had stated that all three CNN architectures achieved broken DI detection and classification accuracy above 0.80 AUC. Lian et al. (Lian et al., 2021) applied a convolutional neural network called nnU-Net to detect caries lesions using deep learning methods, classify different radiographic extensions in panoramic films, and detect caries lesions in their study, in which they aimed to compare the classification results with the results of specialist dentists. In addition, they have applied Dense Net 121 to classify caries lesions according to their depth (dentin lesions in the outer, middle, or inner third D1/2/3 of the dentin). Aminoshariae et al. (Aminoshariae et al., 2021) reported that artificial intelligence can be useful in the detection of periapical lesions and root fractures, evaluation of root canal system anatomy, determination of working length measurements, prediction of tooth pulp viability and success of retreatment procedures. Shagiri et al. (Saghiri et al., 2012) tried to determine root canal lengths with files on radiographic images with and without ANN. According to the results of the measurements made using stereomicroscopy before and after the extraction of the teeth, the precision criterion was found to be 76% for endodontists and 96% for ANN. Qiao et al. (Qiao et al., 2020) used NN to measure root canal length using multi-frequency impedance. It was stated that the NN-based multi-frequency impedance method offers an accuracy rate of approximately 95%.

According to Fukuda et al. CNN-based deep learning model was proposed in the study (Fukuda et al., 2020) for the detection of vertical root fracture on panoramic radiography. The diagnostic performance was found to be 0.75, 0.93 and 0.83 for recall, precision, and F-measure criteria, respectively.

Lee et al. (Lee et al., 2018a) used a pre-trained GoogLeNet Inception v3 CNN network architecture to detect caries in premolars and molars using periapical radiographs. As a result of the study carried out with this CNN-based approach, 89% accuracy value was obtained. Hiraiwa et al. the diagnostic performance of the deep learning system was examined in the study (Hiraiwa et al., 2019), in which the aim was to classify the root morphology of the mandibular first molar teeth on panoramic radiographs. The presented methodology was reported to have a diagnostic accuracy of 86.9% for the interpretation of distal roots. A transfer learning-based Siamese Network was developed in the study of Calazans et al. (Calazans et al., 2022) targeting automatic detection of endodontic lesions by means of cone beam computed tomography (CBCT). It is stated that the proposed classification model produces an accuracy of approximately 70% for the entire image set. In the study by Sajad et al. (Sajad et al., 2019), experiments were conducted on extracting features with AlexNet transfer learning architecture and classifying them with KNN and SVM algorithms for automatic detection of lesion type using periapical X-ray images. As a result of the studies, it was reported to reach a maximum accuracy rate of 98%.

In this study, real-time detection of lesions occurring in the tooth root with a computer-aided system on radiographic images obtained from dental regions using periapical X-ray technique was planned. Due to the fact that X-ray images have a black and white colour format, their interpretation requires clinical experience. On the other hand, regardless of the goodness of image quality and clarity, the interpretation of films varies depending on the mood, experience and even fatigue of the professional. As a result of this change, it is likely that there may be subjective and human-caused errors. It is possible that the X-ray process will need to be repeated due to factors that negatively affect the interpretability of the image. However, minimising the radiation that a person is exposed to during the X-ray shooting process is also a sensitive issue and is the responsibility of the physician. On the other hand, considering the physiological and mental state of the child, the act of re-shooting may need to be repeated many times. This also applies for individuals with disabilities. With this study, the provision of artificial intelligence-supported condition management will make a significant contribution to the world of dentistry. On the other hand, strengthening the critical area in a successfully captured image will contribute to the physician's interpretations.

The methodological diagram, in which the problems and solutions related to the developed system are highlighted as a whole, is given in Figure 1.

In the flow diagram given in Figure 1, an adaptive image processing-based real-time system is proposed for revealing strategic points on the image data obtained from periapical radiographs.

This real-time system offers the potential to support the decision-making processes of physicians for the detection of lesion types such as Primary Endo with Secondary Perio, Primary Endodontic Lesion, Primary Perio with Secondary Endo, Primary Periodontal Lesion and True Combined Lesions.

2. Methodology

Radiographic examination is a preferred method in clinical evaluations performed for the examination of dental diseases. However, in dental and alveolar bone disorders, a mineral density of less than 40% in the relevant tissue reduces the likelihood of dental abnormalities being seen on radiographs. Therefore, providing a computer-aided diagnosis within the framework of image processing and artificial intelligence approach is necessary for the development and standardisation of traditional methodologies (Chauhan et al., 2023).

In this section, an original methodology is presented for the realisation of a successful classification functionality for different types of lesions located in the dental region. In this direction, the format of the dataset used in the study, the original image processing approach and the classification method are described below.

2.1. Dataset

In this study, the free and publicly available Kaggle dataset was used (Periapical X-rays Dataset, 2021). The publicly available dataset named "Periapical X-rays" contains digitised images of the lesion. These categories are Primary Endo with Secondary Perio, Primary Endodontic, Primary Perio with Secondary Endo, Primary Periodontal and True Combined

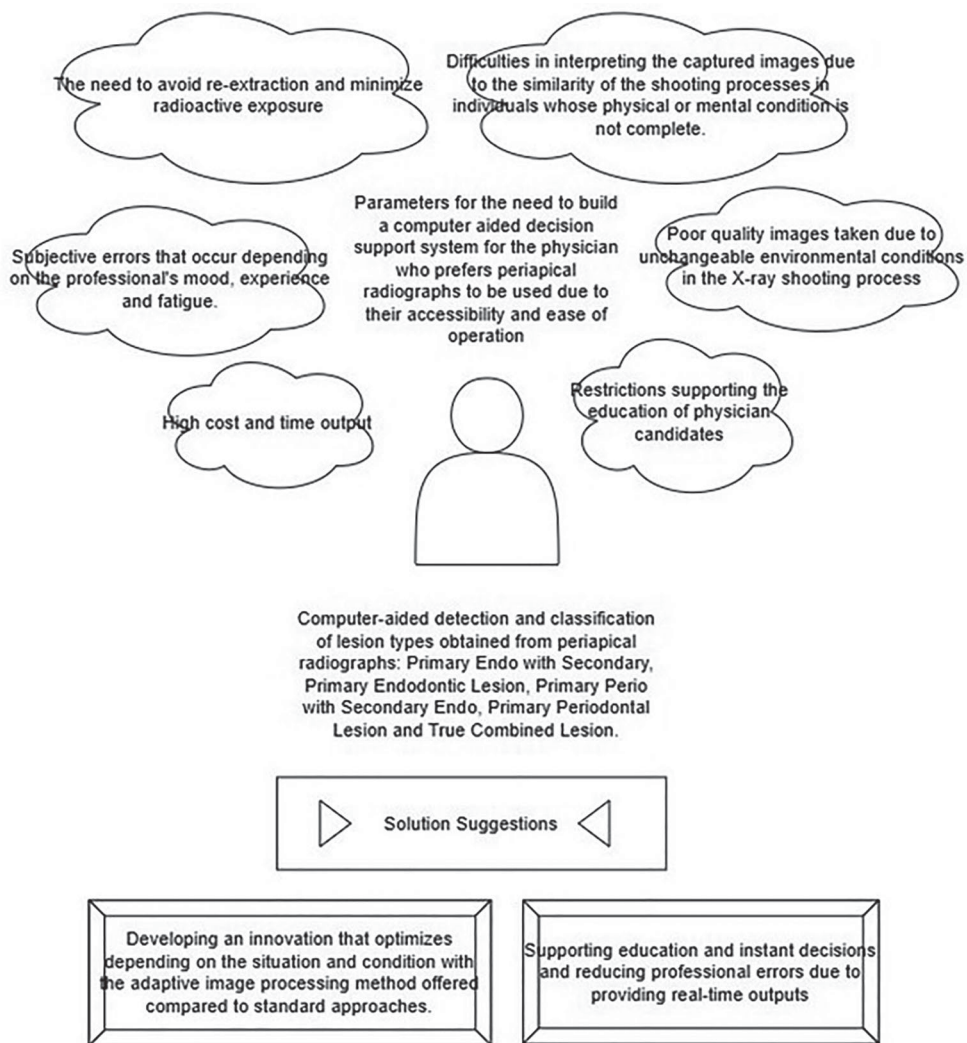


Figure 1. Methodological diagram related to the developed system.

(Sajad et al., 2019). A visual representation of each lesion labelled by an expert radiologist and dentist is given in Figure 2.

The original dataset named "Periapical X-rays" contains 122, 124, 39, 118 and 131 image data, respectively. In this dataset, which has an unbalanced distribution between classes, the representation of each lesion type has been diversified and multiplied by the data augmentation approach. In the scope of the data augmentation approach, horizontal and vertical image translation, random rotation of the image, applying brightness, contrast and colour changes, random zoom-in/-out of the image and data expansion techniques were used. For each category, the original data was increased 3 times to obtain an enriched dataset.

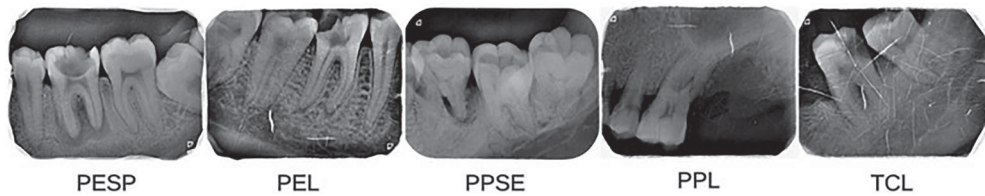


Figure 2. Primary Endo with Secondary Perio (PESP), Primary Endodontic Lesion (PEL), Primary Perio with Secondary Endo (PPSE), Primary Periodontal Lesion (PPL) and True Combined Lesions (TCL) lesion types (Periapical X-rays Dataset, 2021).

In addition, the segmentation process plays an important role in the realisation of the educational process related to object detection. Therefore, the images segmented as a whole by the makesense.ai programme were labelled according to lesion categories. At the end of this process, which was carried out meticulously on the original and enriched datasets, 70% of the datasets were divided into training set, and 30% were divided into test set. Then, the YOLO object detection model for detection and classification of lesion types was trained for two separate datasets via Google Colab with GPU feature enabled.

2.2. Model development

This study is based on the hierarchy of primary processing procedure, evaluation process and model interpretation, which includes the creation of a final prediction model based on the collected dataset regarding the types of lesions.

The primary processing procedure is to classify the dataset using an object detection algorithm after developing it with an image processing approach. The image processing approach used in this section is to perform numerical operations on an image that has been converted to a digital format. This approach allows valuable information to be obtained through mathematical operations and transformations applied to existing images. Mathematical operations performed on images in matrix format are the basis of this approach. As a result of evaluating matrix structures together, image processing techniques emerged.

Image processing techniques are of great importance in terms of understanding and interpreting the problem. Because image data expressed in digital media are 2-dimensional or 3-dimensional matrices ranging from 0 to 255 pixel values. 2-dimensional matrices represent black-and-white images, while 3-dimensional matrices represent colour images. In the black-and-white colour space, a value of 255 pixels corresponds to the colour white, and a value of 0 pixels corresponds to the colour black. Other pixel values between 0 and 255 produce intermediate tones between black and white (Ümit Aksoylu, 2021). The colour space used for colour images is expressed by a 3-channel image matrix. In this direction, the red, green and blue channels combined in different ratios for the target pixel create various colour tones (Hyttinen et al., 2020; Ümit Aksoylu, 2021).

The basic functionalities applied on images consisting of matrices can be listed as brightness and contrast settings, colour correction and balancing, scaling and rotation, noise reduction, edge detection and boundary determination, filters and masking, feature extraction, morphological operations, sensitisation and optimisation (Gonzalez & Woods, 2014; Ümit Aksoylu, 2021). But the operations on the image are not limited to numerical operations only. Artificial neural networks can be used integrated with various subject areas

such as machine learning algorithms and algorithmic techniques. Artificial intelligence techniques used in combination with image processing constitute the computer vision approach (Ümit Aksoylu, 2021).

Image processing and artificial intelligence represent a hierarchical order for the prediction process, which is the output of the primary processing procedure stage. The inferences made by the intuitive mechanisms developed in this layout on the numerical outputs form a prediction at the evaluation and interpretation stage (Ümit Aksoylu, 2021). In this study, it was ensured that the prediction process was optimised with proposed original image processing approach developed to ensure successful diagnoses of the types of lesions related to the dental area.

2.3. Original image processing approach

It is important to evaluate the analytical efficiency using pre-processing techniques on the images. This process allows for the improvement to be observed in the classification accuracy for defective images (Wonggasem et al., 2024), as well as the disclosure of details that cannot be detected with the naked eye on perfect images (Akalin & Yumuşak, 2024). Regarding this situation, it has been stated in the (Akalin & Yumuşak, 2024; Fatima et al., 2023) articles that the image processing approach enhances the performance in the accuracy of detection related to predictions. In this study, an original image processing hierarchy depicted in Figure 3 was proposed to ensure a successful detection of lesion diagnosis.

In the scope of the hierarchy given in Figure 3, the adaptive original image processing approach applied to improve image quality, remove parasites from the image and expose lesion areas is explained step by step with the pseudo-code given in Algorithm 1.

The pseudo-code given in Algorithm 1 describes the operations performed on digital images consisting of pixels. Colour images in digital media have a 3-channel structure. However, due to target-oriented requirements, it is possible to perform many mathematical operations on the image. In this context, in the original image, which is first converted into grayscale space, enabled the necessary preliminary preparation for the reduction of calculation costs and thresholding processes. Secondly, the opening process is performed. The opening process involves the sequential application of dilation and erosion processes. The 3×3 kernel of 1 values interacts with each pixel in the image to provide a matrix update for the output image. With this stage, the noise in the image data is reduced. In the third stage, the thresholding process is performed with the ABC optimisation algorithm, taking into account the qualities that each image has. The thresholding process is an important functionality for the separation of background and objects. Having this functionality in an adaptive format allows the parameters of the shooting process to be adjusted specifically for the image. In the fourth stage, the image is adaptively blurred again based on the qualities that the image has. It offers a critical approach to reducing noise and simplifying details. In the fifth stage, the original, binary and blurred images are combined to ensure that the advantages of each image acquisition process are collected in a single image. In the sixth stage, histogram equalisation is performed to improve visual quality. This technique, which allows pixel intensities to be reconstructed using the cumulative distribution function, is important in the interpretation of X-ray images as it expands the brightness range. In the seventh stage, the reverse of the opening process performed in the second stage is applied. This part is used to combine objects close to each other or to complete missing parts from

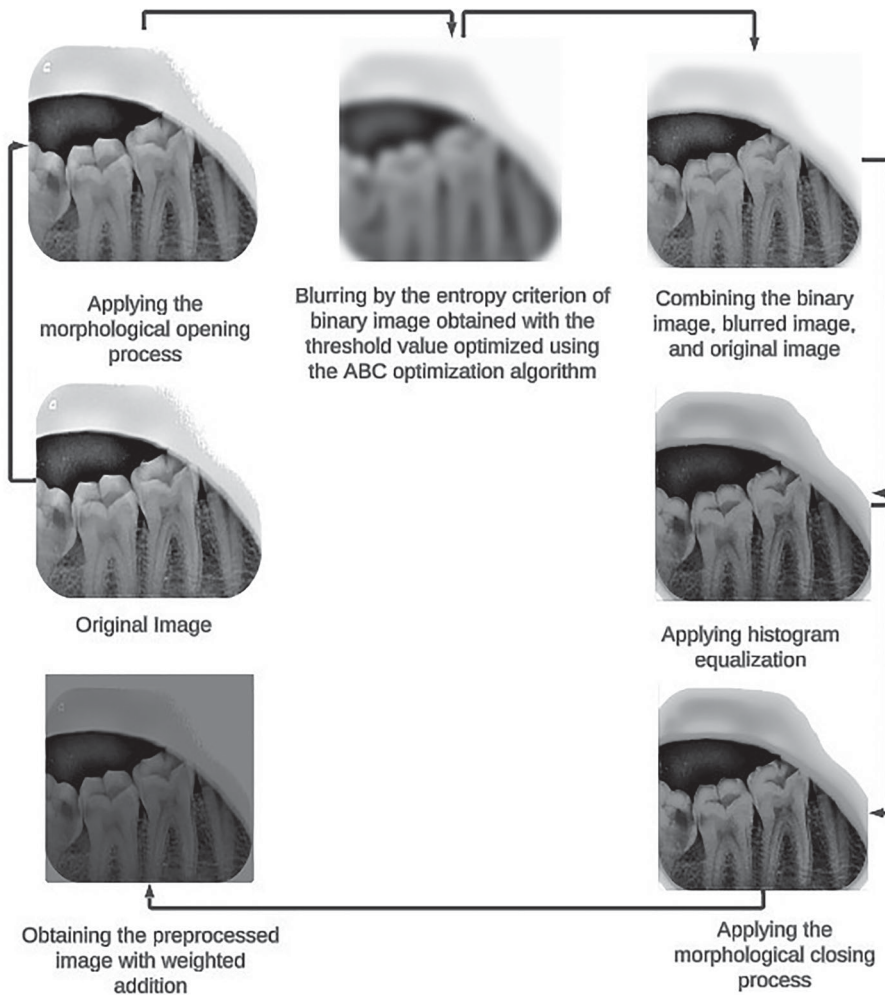


Figure 3. Improved preprocessing hierarchy (Periapical X-rays Dataset, 2021).

a whole. In the last part, weighted computation is made. With weighted summation, it is possible to decide what weight the two images to be merged will have in the final output. It is a valuable step in optimising the image.

In this process, which combines image processing techniques and optimisation, the features of each image are evaluated on the image scope. Then, the images improved with this approach, which offers a unique and adaptable methodology, are classified with the YOLO object detection algorithms, which is stated to optimise the number of parameters and the number of calculations (Wang et al., 2024).

2.4. Feature extraction and classification

In this study, the YOLO framework is used for the feature extraction and classification model. YOLO is a deep learning-based object detection algorithm. It has an important functionality

Algorithm 1. Pseudo-code of proposed image processing approach.

- Start.
 - Take the image.
 - Apply a grayscale filter to the original image and convert it to grayscale image format.
 - (If the image has 3 channels, it is reduced to 2 channels.)
 - Apply the opening process within the scope of the morphological image processing approach to clean 2-dimensional images in grayscale image format from noise.
 - Run the ABC optimisation algorithm for each image obtained by the opening process and determine an appropriate threshold value. During the process of determining the threshold value,
 - 1) Get the dimension information of grayscale images.
 - 2) Create 20 bee objects.
 - 3) Return as many as the number of bees in a loop and update the fitness value with the new threshold value based on the random value assigned to each bee.
 - 4) Create a new scout bee and replace it with the worst bee.
 - 5) Apply morphological opening to the binarized image with the best threshold value determined at the end of the cycle.
 - 6) Return the best threshold value.
 - Assign pixel values less than the adaptively calculated threshold value to 0 for each image. Then assign pixel values greater than the adaptively calculated threshold value to 255 for each image. Thus, get binary images as a result of the assignment process made specific to each image.
 - Apply the Gaussian blur filter, whose blurring parameter is determined by the entropy value of the image, to the original images. (In this section, a relationship is established between the entropy of the image data and the grayscale level by means of an information measure that measures the level of uncertainty of the image. Thus, data-specific inferences are produced by blurring with the image-specific blur parameter.)
 - Perform a merge operation for original images, binary images, and blurred images.
 - If the pixel value of the binary image in the merging process is 0, take the pixel value of the new image to be created from the original image. Otherwise, take the pixel value of the new image from the blurred image. In the final stage, convert to the 8-bit signed integer format so that the range of numeric values related to the merged image is stored in a specific data type.
 - Equalize pixel value distributions by applying the histogram equalisation process to the merged image. (Thanks to the histogram equalisation method, improvement is provided for images that offer low contrast or a wide contrast range.)
 - Ensure that the image is cleaned of external parasites by applying a closing process within the scope of morphological image processing approaches to the image to which histogram equalisation has been applied.
 - Perform weighted summation on the original image with the image to which morphological closure process has been applied. (In this collection process, the weight of the original image on the final output is determined as 0.5, while the effect of the image which closure process has been applied to the final output is included in the processing by taking the negative of the image. Thus, contrast is regulated and certain features are emphasised.)
 - Finish.
-

in computer vision applications for the identification and localisation of objects in images and video (Vijayakumar & Vairavasundaram, 2024).

The YOLO architecture consists of backbone, neck and head components. As a result of modifications made to these components for different versions of YOLO, the overall accuracy and speed of the network are significantly affected.

In the general framework of YOLO, the backbone component is the part responsible for extracting important features from the initial image and creating feature maps. VGG16, ResNet50, CSPDarknet53 and EfficientRep are among the commonly used backbones. The neck component is the part between the backbone and the head in architecture. The main functions of this component are to combine feature maps from the various layers of the backbone network and forward them to the head. Spatial pyramid pool (SPP), Feature Pyramid Network (FPN), NAS-FPN and Rep-PAN are popular YOLO neck options. The head component processes the combined features. In this section, predictions are made about boundary boxes, object probability scores and classification scores. Its main role is to

create the final output of the network. Efficient decoupled, Multi-scale and Anchor-based are among the popular YOLO head components (Ragab et al., 2024).

YOLOv1 is designed to directly predict object locations and classes with a single CNN structure. However, this approach negatively affected the accuracy for small objects and overlapping bounding boxes. YOLOv2, which appeared in this context, has improved object localisation. For this, batch normalisation, anchor boxes and passthrough layers have been introduced. Thus, accuracy and speed have been improved in object detection tasks. YOLOv3 has been developed to include a multi-scale feature extraction architecture. As a result, a balance between speed and accuracy has been achieved. YOLOv4 and YOLOv5 have introduced new network backbones, improved data augmentation and optimised training strategies. These innovations have been reported to improve accuracy. YOLOv6 has implemented the EfficientNet-L2 used in YOLOv5. In addition, the EfficientRep backbone and Rep-Pan neck have been introduced. YOLOv7 has introduced a faster and more powerful network architecture. Thus, it has been provided to improve accuracy through more accurate detection performance, a more robust loss function, and improved label assignment and model training efficiency. YOLOv8 has provided a successful object detection process by offering a mish activation function that increases the processing speed with better feature consolidation and detection accuracy (Ragab et al., 2024). YOLOv9 is a competitive solution with an excellent design that reduces the number of parameters by 49% and reduces the amount of calculations by 43% compared to YOLOv8, resulting in a 0.6% AP improvement. Designed by combining PGI and GELAN neural networks, YOLOv9 offers a strong and stable performance (Wang et al., 2024).

The concept of programmable gradient information (PGI) has been proposed in order to achieve multiple goals in the YOLOv9 architecture. Thus, it is aimed to obtain reliable gradient information for updating network weights. Also, the GELAN network has been developed to provide gradient information to update network weights. In addition to using GELAN and PGI structures together in the YOLOv9 architecture, it has been stated that a superior success has been achieved from pre-trained state-of-the-art models by simplifying the subsampling module CSP-DOWN to the DOWN module (Wang et al., 2024).

2.5. Segmentation and assessment

YOLO is a deep learning-based object detection and classification approach. It performs an object detection operation with a regression problem for class labels determined by means of bounding boxes. In the first stage, the network divides the input image into the SXS grid to predict the final bounding box. Each of these grid cells is responsible for the detection of the object. Therefore, a confidence score is assigned for each cell. Then the confidence score presents the degree of closeness of the box drawn by the model to the true value. In this context, if the object to be detected is not in the box drawn, the confidence score is 0. Otherwise, it is expected to correspond to the intersection over union (IOU) between the estimated box and the ground truth (Redmon et al., 2016). The description of this process is based on 4 basic facts. These phenomena are object localisation, object classification, object detection and object recognition, respectively (Vijayakumar & Vairavasundaram, 2024).

1. Object localisation: The location of each object in the image or video is determined by means of bounding boxes (Vijayakumar & Vairavasundaram, 2024).

2. Object classification: The label assignment of each detected object is performed (Vijayakumar & Vairavasundaram, 2024).
3. Object detection: Objects in video and images are detected and localised by combining object localisation and object classification functionalities (Vijayakumar & Vairavasundaram, 2024).
4. Object recognition: By finding objects on the input image given to the model and then assigning a label to the class object, the probability of an object recognised in the class is produced. (Vijayakumar & Vairavasundaram, 2024).

The performance of object detection models, trained based on the four fundamental principles of the YOLO algorithm, was evaluated using performance metrics such as Average Precision (AP), Mean Average Precision (mAP), Intersection over Union (IoU), recall, precision, and F-score as detailed in (Vijayakumar & Vairavasundaram, 2024) and (Akalin & Yumuşak, 2022b) articles.

Average Precision (AP): Average Precision is the area calculated under the recall and precision curve. This calculation is performed for each of the defined categories.

Mean Average Precision (mAP): It is preferred to evaluate the accuracy and effectiveness of object detection models developed through deep learning techniques. It presents the average of the individual mean accuracy scores of each object class included in the dataset.

Intersection over Union (IoU): It is the process of determining the location errors in the trained object detection models over the correct positions of objects. Within the scope of this method, the overlap between the predicted bounding boxes created by the object detection algorithm and the actual bounding boxes of the objects in the dataset is measured first. Then the intersection of the area between the estimated bounding box and the actual bounding box is divided by the total area to obtain the IoU. A high value of this value means an increase in the quality of the estimate. The IoU is important for the dataset with different object sizes and complexities. It is critical for the reliability of the study that the mAP and IoU values are 100% or close to 100%.

Sensitivity: It is a criterion that defines the power of the trained model to detect real positives. Its mathematical expression is given in equation 1 (Akalin & Yumuşak, 2022b).

$$Sensitivity = TP / (TP + FN) \quad (1)$$

It is aimed that the criterion given in equation 1 is 100% or close to 100%.

Precision: It is a measure that shows the accuracy of the trained model in identifying true positives. Its mathematical expression is given in equation 2 (Akalin & Yumuşak, 2022b).

$$Precision = TP / (TP + FP) \quad (2)$$

It is aimed that the criterion given in equation 2 is 100% or close to 100%.

F Criterion: It is the harmonic mean of the precision and sensitivity values. It ensures that extreme values are taken into account. Its mathematical expression is given in equation 3 (Akalin & Yumuşak, 2022b).

$$F \text{ Criterion} = (2 * precision * sensitivity) / (precision + sensitivity) \quad (3)$$

It is aimed that the criterion given in equation 3 is 100% or close to 100%.

3. Results and discussion

In today's world, computer technology and computer science have the functionality of making sense of the target data by providing inferences through intuitive mechanisms that work on images. This makes it easier to take action on the meaningful data. However, in order to design a system that can behave like a human or have intuitions on images, it is necessary to examine the mathematical basis of the image data to be integrated into the algorithms to be developed and to establish its relationship with the human visual system. In order to establish a strong relationship, as suggested in articles (Andrzejczyk et al., 2024; Kuwil, 2022; Mumtaz et al., 2023; Ramasubramanian et al., 2015), optimisation should be provided on the image by developing original mathematical functions and approaches that will directly affect the pixel values. The development of original functions and approaches is an important issue for image processing. In this context, the clarity of the images was improved by the entropy curve modification performed on the pixel values of the image in the study (Singh Yadav et al., 2024) and successful analysis results were obtained. With the feature extraction approach proposed in the study (Kuwil, 2022), models have been produced in which the distinction between classes is made well. In the study (Sonti & Dhuli, 2022), it was reported that classification performance was improved with a new model presented based on local directional texture descriptor and statistical texture features. In the study (Abbood et al., 2022), an algorithm was developed to improve image quality. The obtained results have shown that the developed algorithm produces a more effective output on classification performance. In addition to the pre-processing steps performed on the image, feature extraction and statistical analysis are other numerical operations. In this regard, it was stated that promising classification results were obtained in the study (Negm et al., 2018), in which important features extracted from the image (shape, geometric, statistical and discrete cosine transform, etc.) were classified. In fact, preprocessing, feature extraction, or statistical analysis approaches performed on the image enable the production of strong inferences by performing successful feature engineering. For this reason, optimising the image by developing original functions or approaches will contribute to the construction of a strong decision support system.

In this study, an original image processing approach was proposed for the detection of 5 different types of lesions that show formation in the root of the tooth on radiographic images obtained from dental areas using the periapical X-ray technique. Then, in order to measure the success of the proposed approach, the original images and the images improved by the image processing approach were classified by the YOLOv9 object detection algorithm, which has a lower computational cost and number of parameters. The classification results on the original and preprocessed images were compared via the Precision-Recall and F1-Confidence curves using the graphs given in Figures 4–7 for 50 and 75 iterations, respectively.

The precision-recall curve allows the precision and recall criteria of the model to be evaluated together on the same graph. The results with a high precision and recall ratio on the precision-recall curve plot indicate that the model produces less false positive predictions and true positives have been successfully detected. The parts with low precision and high recall rates indicate that incorrect diagnoses were made for most of the positive predictions produced by the model. The High Precision and Low Recall regions, on the other hand, explain the situation in which the true number of positive samples is estimated by a small

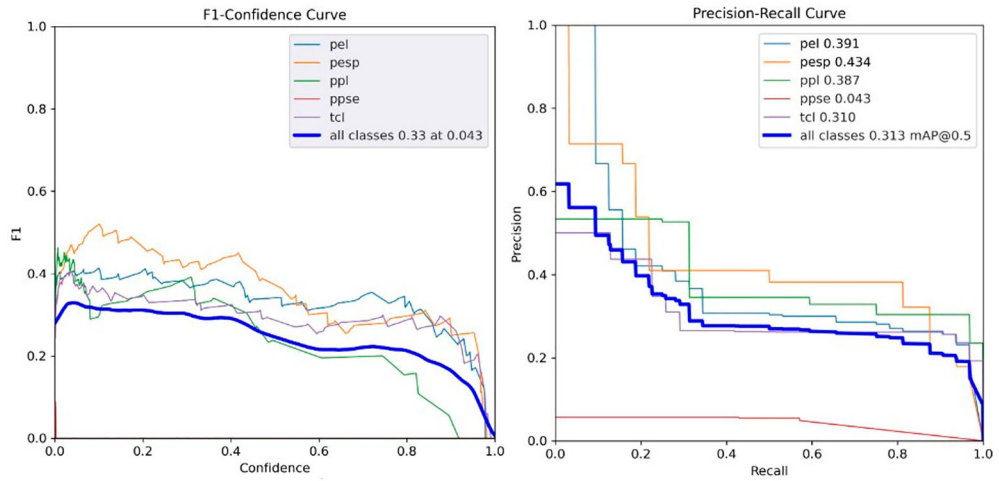


Figure 4. Performance metrics obtained after training YOLOv9 with the original images for 50 iterations.

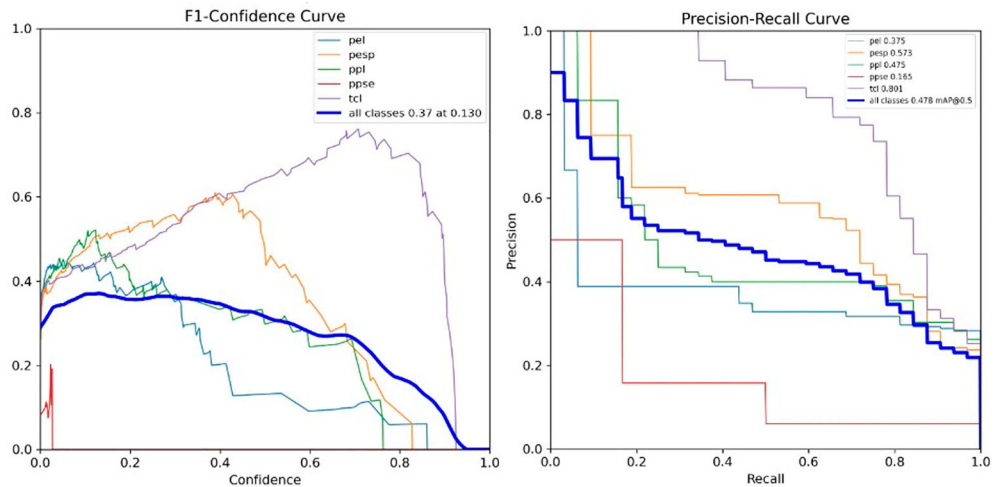


Figure 5. Performance metrics obtained after training YOLOv9 with the enhanced images for 50 iterations.

amount. Moreover, it is also important to interpret the area under the precision-recall curve (AUC-PR), and it takes a value between 0 and 1. A value close to 1 indicates that the model is performing very well, while a value close to 0 describes that the model is performing poorly.

In the light of this information, the performance curves produced for the detection of Primary Endo with Secondary Perio (PESP), Primary Endodontic Lesion (PEL), Primary Perio with Secondary Endo (PPSE), Primary Periodontal Lesion (PPL) and True Combined Lesions (TCL) lesion types were compared for the original and enhanced images. In the training performed for 50 iterations, the performance measured in the general precision-recall and F1 criterion curves was higher on the enhanced images. An overall improvement was also achieved in the detection accuracy for each category. It is important to achieve the maximum result by changing the number of iterations, which is a hyperparameter.

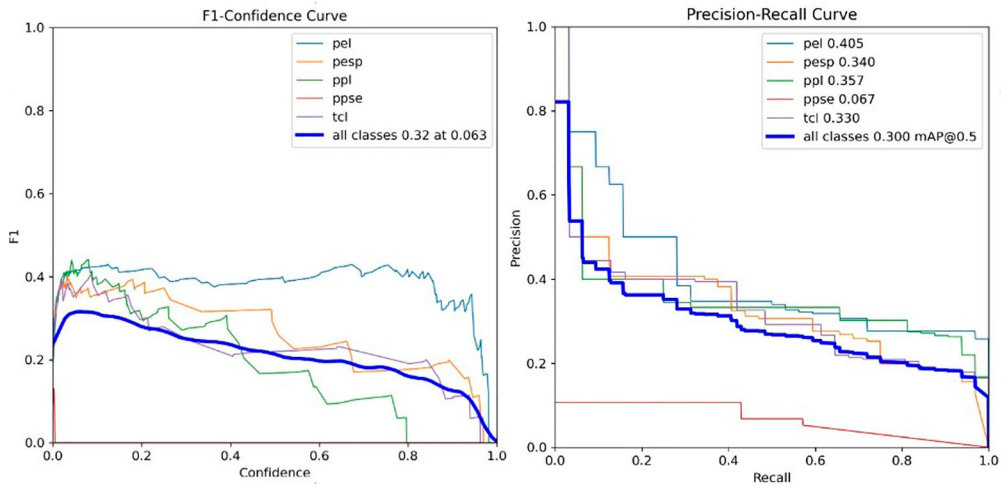


Figure 6. Performance metrics obtained after training YOLOv9 with the original images for 75 iterations.

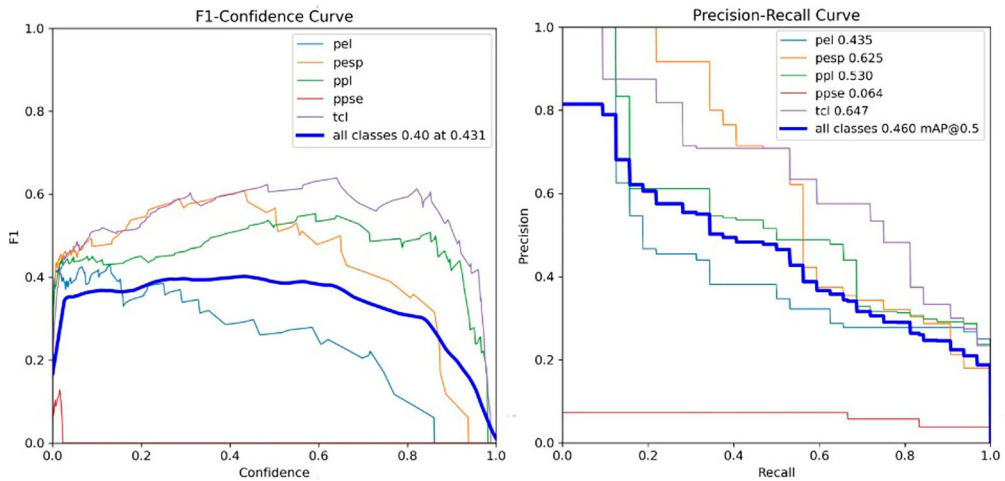


Figure 7. Performance metrics obtained after training YOLOv9 with the enhanced images for 75 iterations.

Therefore, retraining was performed for 75 iteration values. When the performance outputs obtained for the original and enhanced images as a result of training are examined, it is seen that the images preprocessed with the proposed image processing approach produce a more successful output both on a general and categorical basis. During the training process of the YOLO v9 architecture, the batch value was determined to 16. The default values of the other hyperparameters were used.

After deciding that the preprocessed images were more successful compared to the original images, the outputs were evaluated for the number of iterations. In this direction, for the preprocessed images, the overall accuracy in the Precision-Recall curve of the training performed with 50 iterations is 0.03 higher than the training performed with 75 iterations (Figure 5 and Figure 7). However, the f-criterion, which reduces the balance between

precision and recall to a single number, produces a more successful overall result for training over 75 iterations. In addition, obtaining similar rates of detection accuracy in each category for the 75 iterations training is an important step in determining the number of iterations. On the other hand, increasing the number of iterations will cause the optimum performance value to decrease slowly, as seen in the graphs presented in Figure 5 and Figure 7. Therefore, the number of iterations was decided as 75. Also, the recognition rate for the TCL category in each of these charts is at very low levels. This is because, among the 5 basic lesion types, the number of samples in the TCL lesion type is almost 1/3 compared to the number of samples presented for other lesion types.

In summary, it is seen that the improved images as depicted in the graphics offer a more successful and stable result in model training compared to the original images. This is explained by cleaning the original images from the parasites thanks to the unique image processing approach and emphasising the critical regions in the images with adaptive methods specially developed for each image. Because each image contains environmental factors that vary. These factors have consequences that affect each image specifically, such as noise, image contrast, blur level, composition and angle. For this reason, sensitive, critical, ambiguous, and important details have been provided to reveal with the adaptive image-processing approach developed specifically for the image. This functionality is also important to strengthen the manual interpretability of information in the images and to provide better input for artificial intelligence-supported automatic inference systems (Maini & Aggarwal, 2010).

The techniques developed within the scope of image processing techniques without distorting the digital image provide convenience in making sense of the process. In this context, when the graphs given in Figures 4–7 are evaluated, it can be interpreted that the model training performed on the images improved by the proposed image processing approach increases the diagnostic success. But despite this, overall performance is low. This is explained by the insufficient number of data and the unbalanced distribution of data between categories. Because in deep learning-based studies, appropriate data is the basis of the learning functionality. Among the most important factors determining the overall performance and accuracy of the algorithm is the availability of a quality, diverse and balanced dataset. Insufficient data and unbalanced data distribution between classes negatively affect training performance for deep learning-based approaches that have the potential to achieve success through data. Therefore, the data augmentation approach has been used to improve training performance.

Data augmentation is a synthetic data generation process performed through image data in an existing dataset to increase the generalisation ability of the model and the ability to adapt to different variations. After this process, the diversified and growing dataset may consist of small modifications made to the original data, or it may also contain combinations between remote samples that would be difficult to extract (Shorten et al., 2021).

The diversified and growing dataset after the synthetic data generation approach is critical for training deep neural network-based approaches. In particular, the prevention of excessive compliance finds a solution as a result of the data augmentation approach. Otherwise, deep neural network-based approaches used for insufficient and unbalanced datasets cause the model to tend to learn false correlations (Shorten et al., 2021).

In this study, processed images with higher performance compared to the original images were diversified and augmented with a data augmentation approach. In the scope

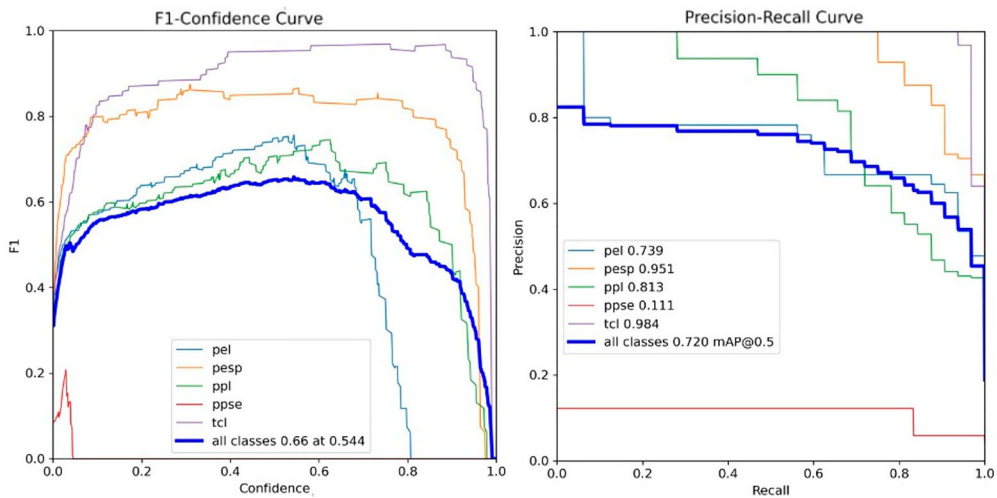


Figure 8. Performance metrics obtained by training the YOLOv7 neural network for 75 iterations on the dataset enriched with enhanced images.

of this approach, horizontal and vertical image flipping, random rotation of the image, random cropping of the image provided that the lesion areas are protected, implementing brightness, contrast and colour changes, random zoom-in/-out of the image and data expansion techniques have been applied. In the dataset in which the number of samples is increased by using these techniques, the replication process is shaped by increasing each category by 3 times. As a result, image data in the TCL category constitutes a smaller proportion of the dataset compared to other categories. Then, the current dataset constructed was divided into training and test sets at 70% and 30% rates, respectively. At the last stage, the augmented training dataset containing sufficient data was trained with the YOLOv9 architecture. YOLOv9 is a preferred network because it is efficient in terms of computational cost and number of parameters. However, in the medical field, performance has a higher importance when the relationship between the computational cost of the network and the success of the network is taken into account. Therefore, the augmented dataset has also been trained on the lightweight architectures of the YOLOv7, YOLOv8, and YOLOv10 versions.

As a result of training, the test data given as input to 4 different neural networks were trained for 75 iterations. The batch value of the YOLO architectures was determined as 16 during the training process. The default values of the other hyperparameters were used. The performance values achieved for 4 separate YOLO versions are shown in the graphs presented in Figures 8–11.

In the graphs shown in Figures 8–11, as a result of the data augmentation process applied to the images enhanced with the proposed image processing approach, the training performed with the expanded and diversified dataset reached a satisfactory performance level with the YOLOv8 neural network method. This situation changes due to the learning mechanism of the network built within the scope of YOLO versions, feature extraction ability, complexity and capacity of the model, and optimisation methods performed for the neural network. In this direction, the training strategy of the YOLOv8 architecture

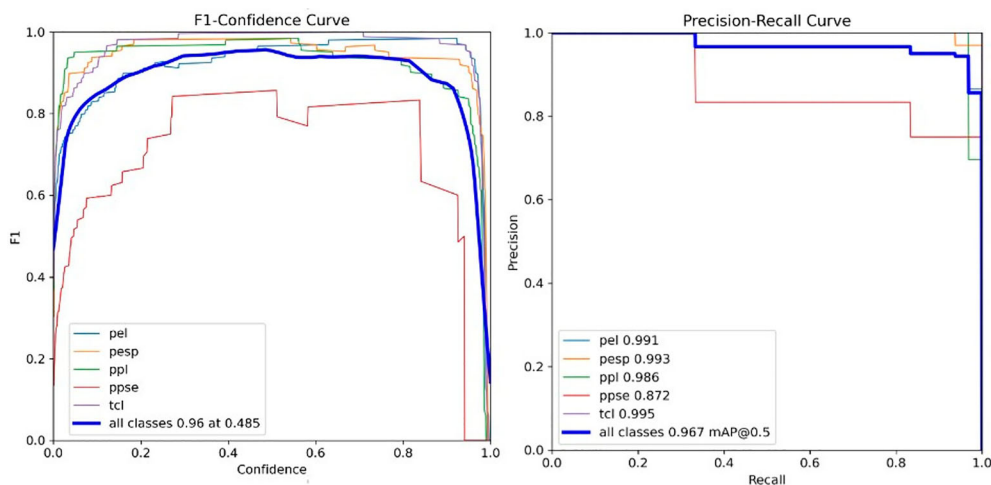


Figure 9. Performance metrics obtained by training the YOLOv8 neural network for 75 iterations on the dataset enriched with enhanced images.

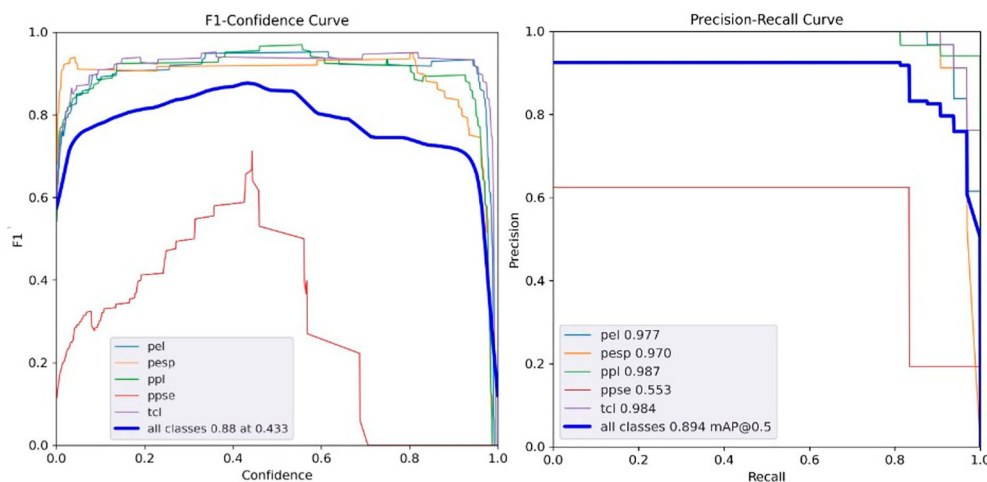


Figure 10. Performance metrics obtained by training the YOLOv9 neural network for 75 iterations on the dataset enriched with enhanced images.

produced an output more appropriate to the type of problem and the dataset. The graphs given in Figures 8–11 confirm this situation. However, among the 5 different lesion detections in the YOLOv8 architecture, the detection success of image data with TCL category is low. This is because each category is increased 3 times within the scope of the data augmentation approach. Therefore, the data of TCL category is lower in the enriched data set than in the other categories, as in the original data set.

Another basis for the success achieved on the enriched dataset of the YOLOv8 approach is that the negative effects of environmental factors such as noise, image contrast, composition, angle, etc. that affect the process of obtaining the images are minimised by the proposed image processing approach. This is very important. Because the periapical

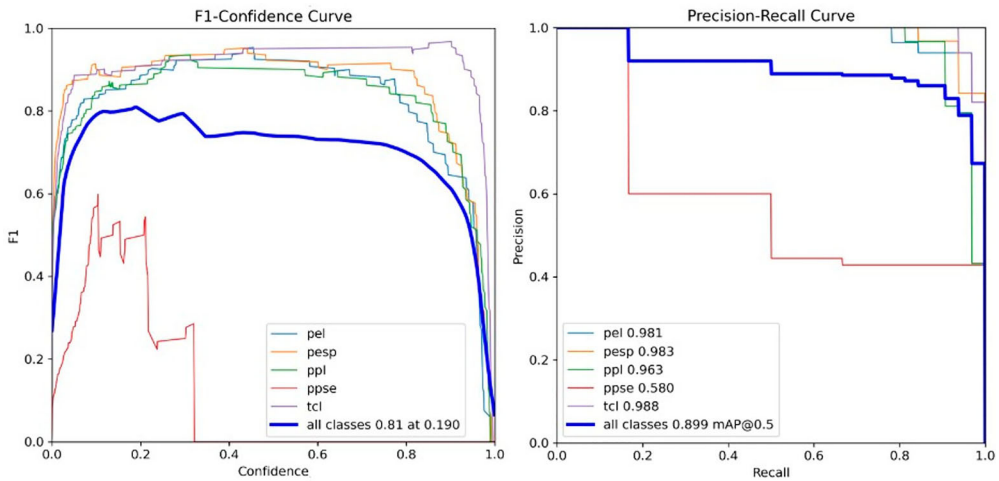


Figure 11. Performance metrics obtained by training the YOLOv10 neural network for 75 iterations on the dataset enriched with enhanced images.

X-ray taking process of individuals with various age distributions differs due to both physical and mental development processes. In such a case, it would not be healthy to make the same inference for each image as a result of applying the uniform image processing approach developed to all images. Therefore, performing operations on image-specific features with adaptively determined parameters will ensure that the potential inference process is evaluated fairly for all images.

This study has a limitation. During the labelling process performed to be given as input to the YOLO frame, coordinate information about the lesion elements on the image is not available. Therefore, the images as a whole have been labelled as PEL, PESp, PPL, PPSE and TCL. This situation negatively affects the detection of where the lesion is located. Therefore, there was no need to use Average Precision (AP) and mean Average Precision (mAP) performance metrics. Because these metrics are valuable for measuring the overall accuracy of bounding boxes. On the other hand, the inference regarding the type of lesion the image has is important as it is useful in determining the type of lesion predicted in the patient. Some deep learning-based studies have been performed in the literature to detect lesions or conditions causing lesions through periapical radiographs are presented in Table 1.

The studies given in Table 1 present a methodology based on the convolutional neural network for the diagnosis and classification of problems in the field of dentistry. The convolutional neural network has made significant contributions to the field of medical science in recent years (Anwar et al., 2018; Sarvamangala & Kulkarni, 2022; Yadav & Jadhav, 2019; Yu et al., 2021). In particular, the use of Convolutional Neural Network in the field of dentistry has been examined in (Lee et al., 2018a; Lee et al., 2018b; Miki et al., 2017; Prajapati et al., 2017; Schwendicke et al., 2019). In the studies (Al Hasan et al., 2023; Chen et al., 2022; Hiraiwa et al., 2019; Wang et al., 2023; Yang et al., 2018), evaluations were made specifically related to the field of endodontics. On the other hand, many deep learning-based studies have been conducted (Calazans et al., 2022; Çelik et al., 2023; Endres et al., 2020; Gao et al., 2024; Liu et al., 2023; Sajad et al., 2019) focusing on the detection of lesions or conditions causing lesions, similar to the subject of this article. When these studies based on deep

Table 1. Reviewing some deep learning-based studies on lesions or conditions causing lesions.

Paper and year	Aim	Methods	Performance Criteria	Research gap future aims
(Sajad et al., 2019)	Automatic detection of lesion type on periapical X-rays	Transfer learning approach and traditional machine learning classifiers	Maximum accuracy rate: 98%	In the future, an idea has not been declared, which is planned to be a continuation of the proposed article.
(Calazans et al., 2022)	Classification of teeth as healthy or with endodontic lesions using cone beam computed tomography images	Innovative transfer learning-based Siamese Network	Approximate accuracy rate for the entire image set: 70%	The proposed Siamese network is planned to be used in different classification tasks.
(Endres et al., 2020)	Detection of radiolucencies using a dataset consisting of panoramic radiographs	Deep convolutional neural network	Average accuracy: 0.60 (± 0.04) F1 score: 0.58 (± 0.04) PPV: 0.67 (± 0.05) TPR: 0.51 (± 0.05)	In the future, it is planned to develop data-fusion models that produce reliable diagnoses by combining continuously learning and evolving models with radiographs.
(Liu et al., 2023)	Detection of periodontitis at an early stage	PAR-CNN architecture based on deep learning	The accuracy rate of the PAR-CNN model for periodontologists and general dentists is 0.800, 0.813, and 0.693, respectively.	In the future, it is planned to include clinical information related to periodontitis, such as blood lipids, rheumatoid arthritis, and cancer, in the study.
(Çelik et al., 2023)	Automatic detection of periapical lesion on panoramic radiographs	Transfer learning based Faster R-CNN, RetinaNet, YOLOv3, SSD, Libra R-CNN, Dynamic R-CNN, Cascade R-CNN, FoveaBox, SABL, ATSS architectures	Maximum accuracy value: 69.2%	It has been stated that the limitations of integrating artificial intelligence applications into clinics should be addressed. It is planned that efficiency in health will be improved by resolving these limits.
(Gao et al., 2024)	Artificial intelligence-assisted diagnosis of dental caries, dental defects, periapical lesions and coronal restorations on periapical films	YOLO-DENTAL network based on YOLOv7-X	The maximum mAP value obtained by the YOLO-DENTAL approach: 86.81%	In the future, it is planned to diversify the dataset for the generalisation ability of the model and to generate a stable output of unified learning techniques on real-world problems to create a generalisable model.
Our proposed study	Detection and diagnosis of endo-perio lesions with periapical radiographs	Classification of the balanced dataset obtained by the data augmentation approach applied to the images improved with the proposed image processing technique on the deep learning-based YOLO architectures	F1 Score: 96%	In the future, it is planned to modify the YOLO architecture to strengthen the feature extraction on images.

learning are examined, it is seen that they are generally built on the convolutional neural network approach that automatically extracts features from images. In this study, based on the same approach, the YOLO object detection algorithm was also used. YOLO, which identifies and locations objects quickly and effectively, is a method that has proven its success in many fields, including the field of medicine (Vijayakumar & Vairavasundaram, 2024). Because it offers a one-stage object detection process, it significantly reduces the computational requirements. Its strong compatibility with industrial demands, lightweight design, and combination of accuracy and speed are its outstanding features. In addition, the flexibility of the model offers a wide range of applications. In fact, since the introduction of the YOLO object detection algorithm, approaches have been developed that address previous versions and enhance overall effectiveness (Safaldin et al., 2024). In this study, training was performed with lightweight architectures of YOLOv7, YOLOv8, YOLOv9 and YOLOv10 neural networks in order to achieve the performance values that produce the most effective output. YOLOv8 has been selected as the architecture that showed maximum success on the enriched and preprocessed dataset with 96% F1 Criterion rate.

In the future study, it is aimed to develop this artificial intelligence-supported system, which is planned to be integrated into the clinic, to facilitate the diagnostic processes of physicians, to minimise technical errors and to prevent complications that may develop due to technical errors. In addition, in patients with difficult cooperation (children, mentally retarded, disabled, etc.) avoiding the repetition of radiography is another advantage. On the other hand, despite the image quality, there is the potential to prevent errors that may arise due to interpreters. Since it is a real-time automatic system, clinical operations will be optimised with detection and classification produced within seconds. In the next study, it is planned to construct the detection and classification process related to lesions with mobile support.

4. Conclusion

In recent years, artificial intelligence has become a reliable technology in clinical decision support systems with the solution proposals it offers in the field of medicine. This success shows that there is a potential for detecting 5 different types of lesions in the tooth root through radiographic images. Deep learning-based approaches are preferred to reveal this potential. Because feature extraction, which exists within the scope of deep learning-based approaches, is an important functionality for the process of making sense of and interpreting images. However, the process of taking periapical radiographs, which are aimed to be analyzed, differs due to the physical and mental development process of individuals with various age distributions. Therefore, it is critical to pre-process the images before applying deep learning-based approaches on radiographic images to maximise performance. For this, an adaptive image processing approach specific to the image was proposed. Thanks to this approach, in which optimisation and image processing methods are used together, it has been ensured that unclear points are illuminated and strategic parts are clarified. Then, the images improved by the adaptive image processing approach were enriched by the data augmentation approach. Finally, these images were trained by the YOLOv7, YOLOv8, YOLOv9 and YOLOv10 neural network architectures based on deep learning. As a result of the training, the improved images compared to the original images reached the 96% F Criterion thanks to the training strategy and network architecture of the YOLOv8 architecture.

This proves that the YOLOv8 algorithm is a more suitable architecture for the problem of classifying lesion types used in the dental field. This proposed methodology is planned to be a decision-making mechanism that supports dentists.

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Author contributions

Fatma AKALIN: Conceptualisation, Data Curation, Formal Analysis, Methodology, Investigation, Resources, Software, Supervision, Validation, Visualisation, Writing – Original Draft, Review and Editing. Tuğçenur YILDIZ: Investigation, Supervision, Data Curation, Validation, Visualisation, Writing – Original Draft, Review and Editing. All authors have read and agreed to the published version of the manuscript.

Data statement

The datasets used during the current study are publicly available on Kaggle: <https://www.kaggle.com/datasets/muhammadsajad/periapical-xrays>.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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